Expertise Validation Form

Automated Viva Assessment Using Multi-Modal Analysis (Video, Audio and Speech), Live Interviewer Copilot, and Large Language Models.

Academic Expert Review and System Validation

1. Expert and Evaluation Details

Lecturer's Full Name: .....

Designation / Academic Position: .....

Faculty / Department: .....

Institution: Sri Lanka Institute of Information Technology (SLIIT)

Date of Evaluation: .....

2. Student Details and Research Details

Student Name – B.K.H.M.B.L. Herath

Student Id – IT22557056

Research Group Id – R26-SE-025

Research Title - AI-Integrated Multi-Modal Academic Evaluation, Performance Intelligence, and Automated Viva Assessment System

3. Purpose of Validation

This document is intended to obtain an independent expert's assessment of the Viva Evaluation System developed for academic assessment, comprising (a) analysis of recorded viva videos and (b) the Live Interviewer Copilot used during live viva sessions. The expert is requested to review the system demonstration and/or sample test cases and evaluate its functionality, accuracy, marking consistency, and suitability for academic use.

The expert is also asked to mark each recorded viva independently using the attached Viva Marking Sheet and grading workbook (Excel), before seeing the system's result, so that the expert's marks can be compared with the system's marks.

Please tick **one option** for each criterion based on the evidence demonstrated. Select "Cannot Validate" if the criterion cannot be confirmed or requires further testing. For any criterion marked "Cannot Validate", please note the criterion number and what is needed in Section 7 (Suggestions and future enhancements). Additional comments and recommendations are welcome.

4. Scoring Metric Used by the System

The official AI performance mark (/100) is the equal-weighted average of the available components below. A component with no usable data is dropped and the rest are renormalised. All behavioural signals listed are scored inside these components. Gaze, head stability and speech emotion are operational proxies and have not yet been validated against labelled viva data; this review is part of that validation.

Component	Metrics	Role in mark
Engagement	CNN frame engagement score; facial confidence; facial emotion profile; on-camera gaze ratio; head stability; blink rate (equal parts)	Scored (1/3)
Audio acoustics	Pitch stability, HNR clarity, jitter and shimmer; emotional tone of voice from speech emotion recognition (half weight, real model only)	Scored (1/3)

Component	Metrics	Role in mark
Transcript	Speech rate band, hedges, fillers, long pauses, sentence completion	Scored (1/3)
Technical accuracy (optional)	Examiner-confirmed score /10; final = 50% AI + 50% technical	Scored when enabled

Quality gates. Before any mark is given, the recording must pass the gates below. A recording that fails a gate is reported as INCOMPLETE with the reason and what to do, and no official mark is given. An incomplete result is not a zero.

Gate	When it fails	Result
Usable face	No face found on camera in enough frames (fewer than 3 frames, or under 15% of sampled frames)	Incomplete
Facing the camera	A face is visible but mostly turned away (non-frontal) so emotion and engagement cannot be measured	Incomplete
Single student	More than one person is clearly visible in frame (a second face of significant size over several frames)	Incomplete
Speech present	The student is on camera but does not speak (no usable words and a silent audio track)	Incomplete
Lip movement	Speech is detected but almost no mouth movement is visible (guards against a still photo with audio), or mouth movement cannot be measured	Incomplete
Audio usable	Audio is shorter than 1 second or empty, or the audio/speech-to-text step failed (reported as a pipeline fault, not a silent student)	Incomplete
Scorable evidence	No component has usable data to score	Incomplete

5. System Validation Checklist

No.	Validation Criterion	Can Validate	Cannot Validate
1	The system accepts a recorded viva video (upload) or a live viva session and processes it successfully.		
2	The system detects the student's face and applies a quality gate; recordings without a usable face are flagged as incomplete rather than marked.		
3	The system detects a face that is mostly turned away from the camera (non-frontal) and flags the recording as incomplete, asking for a re-recording facing the camera.		
4	The system detects more than one person in the frame and flags the recording as incomplete rather than marking it.		
5	The system detects a student who is on camera but silent and flags the recording as incomplete (no official mark) instead of scoring it.		
6	The system checks that mouth movement is visible when speech is detected, so a still photo with audio is not marked.		

No.	Validation Criterion	Can Validate	Cannot Validate
7	The system rejects audio that is too short, empty or failed in processing, and tells a pipeline fault apart from a silent student.		
8	Every incomplete recording shows a clear reason and what the user should do (for example, re-record), and incomplete is not treated as a zero mark.		
9	Dark, blurry or low-contrast faces are enhanced and flagged rather than discarded, and unreliable measurements (for example blink rate on a very short clip, or a short transcript) are flagged as low reliability.		
10	The system recognises facial expression/emotion from the video and reports an emotion profile over time.		
11	The system computes a frame-level engagement score from the student's facial behaviour (CNN engagement model).		
12	The system measures eye gaze (on-camera ratio), head stability and blink rate, and these contribute to the Engagement component of the mark.		
13	The system converts the student's speech to text accurately (audio extraction and speech-to-text).		
14	The system recognises emotion from the student's voice (speech emotion recognition); it contributes to the Audio component at reduced weight.		
15	The system measures voice quality from the audio (pitch stability, clarity/HNR, jitter and shimmer).		
16	The system analyses speech delivery from the transcript (speech rate, long pauses, filler words, hedging, sentence completion).		
17	The system evaluates the relevance of the student's answers to the examiner's questions (Q&A relevance).		
18	The system supports technical-accuracy assessment against the subject rubric or project context, with the examiner confirming the technical score.		
19	The scoring metric in Section 4 is applied correctly: the official mark (/100) is computed from the weighted components, and unavailable components are excluded rather than penalised.		
20	The system shows a per-component score breakdown so the examiner can see how each metric contributed to the final mark.		
21	The system supports the examiner workflow for marks (draft, review and publish); technical-accuracy marks are never published automatically.		
22	The Live Interviewer Copilot transcribes the student's speech in real time during the presentation and viva.		
23	The Live Interviewer Copilot suggests relevant follow-up questions, and the examiner decides which to ask (human-in-the-loop).		
24	The evaluation results and AI-generated feedback are understandable and useful.		

No.	Validation Criterion	Can Validate	Cannot Validate
25	The system produces reasonably consistent results for comparable recordings.		
26	The system's workflow and outputs are suitable for the intended academic users (lecturers and examiners).		
27	Overall, the system demonstrates potential for supporting viva assessments in academic settings.		
28	The system's marks are close to my own independent marks for the same recordings (see the attached grading workbook).		

6. Overall Expert Assessment

Based on the system demonstration and evidence reviewed, please select one option.

[ ] **Can Validate:** The system is suitable for the intended viva evaluation task.

[ ] **Can Validate with Improvements:** The system is suitable after the recommended improvements are completed.

[ ] **Cannot Validate at Present:** The system requires further development and/or testing.

[ ] **Unable to Determine:** There is insufficient evidence to make an overall assessment.

7. Expert Comments, Suggestions and Future Enhancements

Strengths identified:

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Issues identified / Recommended improvements / Suggestions and future enhancements: For every criterion marked "Cannot Validate" in Section 5, please write the criterion number and what is needed (for example more test cases, a feature that is missing, or further development).

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8. Expert Confirmation and Signature

I confirm that I have reviewed the Viva Evaluation System through a demonstration and/or sample test case. The selections and comments recorded in this document represent my professional assessment based on the evidence presented. The expert's name, designation and date are recorded in Section 1.

Signature: .....

Thank you for your valuable expert contribution to this research and system validation process.